Clostridium difficile infection in the community: a zoonotic disease?

Clostridium difficile infections (CDIs) are traditionally seen in elderly and hospitalized patients who have used antibiotic therapy. In the community, CDIs requiring a visit to a general practitioner are increasingly occurring among young and relatively healthy individuals without known predisposing factors. C. difficile is also found as a commensal or pathogen in the intestinal tracts of most mammals, and various birds and reptiles. In the environment, including soil and water, C. difficile may be ubiquitous; however, this is based on limited evidence. Food products such as (processed) meat, fish and vegetables can also contain C. difficile, but studies conducted in Europe report lower prevalence rates than in North America. Absolute counts of toxigenic C. difficile in the environment and food are low, however the exact infectious dose is unknown. To date, direct transmission of C. difficile from animals, food or the environment to humans has not been proven, although similar PCR ribotypes are found. We therefore believe that the overall epidemiology of human CDI is not driven by amplification in animals or other sources. As no outbreaks of CDI have been reported among humans in the community, host factors that increase vulnerability to CDI might be of more importance than increased exposure to C. difficile. Conversely, emerging C. difficile ribotype 078 is found in high numbers in piglets, calves, and their immediate environment. Although there is no direct evidence proving transmission to humans, circumstantial evidence points towards a zoonotic potential of this type. In future emerging PCR ribotypes, zoonotic potential needs to be considered.